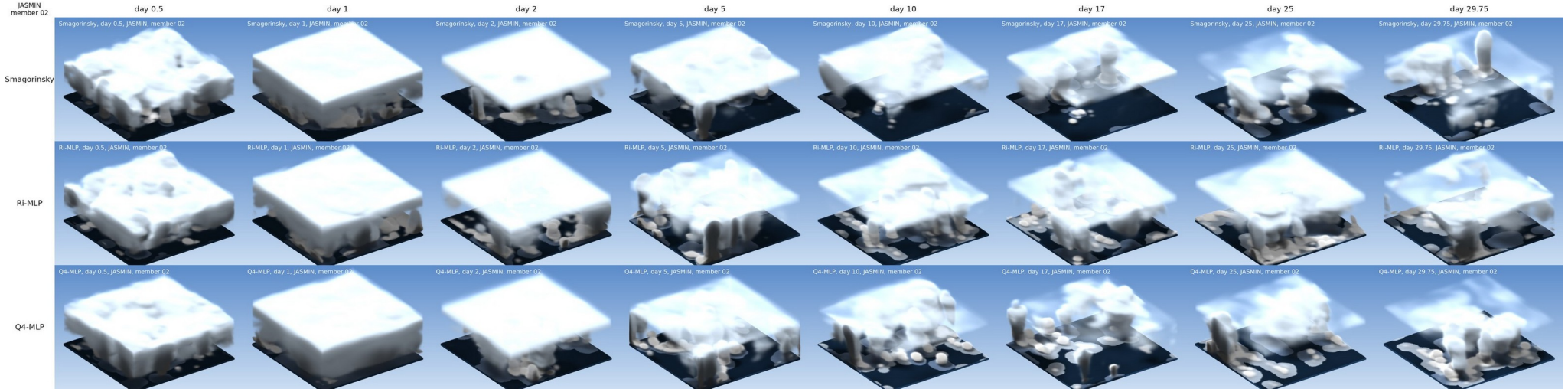
AFESP RESEARCH HIGHLIGHT

# Neural networks that emulate turbulence in the MONC cloud model




*Simulated clouds over a month (left to right) with MONC's original turbulence scheme (top) and the two neural-network emulators; 64 × 64 km domain, 1 km grid.*

**Sambit Kumar Panda** develops neural-network emulators of turbulence in the Met Office NERC Cloud Model (MONC). Using physics-guided multi-task learning, the networks learn MONC's Smagorinsky–Lilly subgrid turbulence scheme with atmospheric stability built in. This year the work was published and presented at **IJCNN 2026** and **PASC '26**. The emulators now run inside MONC in place of the original scheme, and the model stays stable for month-long simulations, including on grids and in warmer climates the networks never saw. His poster at the ICCS Summer School 2026 won the **Best Poster award (runner-up)** and he also presented his work at the km-scale Global Modelling Sumit 2026 at Hamburg this year.

**IJCNN 2026 · IEEE**

Physics-guided multi-task learning for SGS turbulence parameterization

doi.org/10.1109/IJCNN69854.2026.11728747

**PASC 2026 · ACM**

Physics-aware multi-task learning for atmospheric turbulence parameterization

doi.org/10.1145/3815572.3815750

**Best Poster, runner-up**
ICCS Summer School 2026, Cambridge

**PRESENTED AT**

- IJCNN 2026, Maastricht · Oral
- PASC 2026, Bern · Oral
- km-Scale Global Modelling Summit 2026, Hamburg · Poster
- EXCLAIM 2025, ETH Zurich · Poster
- EGU General Assembly 2025 · Oral

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